Sohan Jayram Reddy Kadiri

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PROFESSIONAL SUMMARY

Machine Learning and Data Engineer with experience building ML pipelines, distributed data systems and AI-powered applications using Python, SQL and PyTorch. Skilled in model training, data processing, statistical evaluation and cloud deployment.

EDUCATION

Northeastern University Sep 2025 - May 2027
Master of Science, *Computer Science* GPA: 3.9

Siddaganga Institute of Technology Dec 2021 - Jun 2025
Bachelor of Technology, *Artificial Intelligence and Data Science*

PROJECTS

GPT-2 Optimization — Distributed Language Model Training | Python, PyTorch, DeepSpeed, CUDA, Distributed Training

- Optimized **GPT-2 training using PyTorch and DeepSpeed**, scaling execution across 4 NVIDIA V100 GPUs to improve distributed training performance.
- Increased training throughput by **3.4×**, from 29,600 to 100,791 tokens/sec, scaling from 1 to 4 GPUs in the documented benchmark.
- Optimized **GPU memory utilization and training efficiency**, reducing peak memory consumption by 35% while maintaining less than 1% validation-loss deviation.

Insight — Real-Time Data Processing & Analytics Platform | TypeScript, Kafka, PostgreSQL, Redis, NestJS, Docker

- Engineered a **multi-tenant data ingestion pipeline using Kafka, PostgreSQL and Redis**, processing 41,100 events with zero request errors in a documented load test.
- Implemented **idempotent stream processing and transactional SQL aggregation**, preventing duplicate counts and routing failed events to dead-letter queues.
- Increased processing throughput by **61×**, from 46 to 2,816 events/sec, using Kafka batch consumption, bulk PostgreSQL writes and pipelined Redis operations.

Reddit Sentiment Analysis — NLP Classification Platform | Python, PyTorch, DistilBERT, Streamlit, AWS, Docker

- Built an **end-to-end NLP data pipeline using Python and PRAW**, collecting and preprocessing 10,000+ Reddit posts for sentiment classification.
- Fine-tuned **DistilBERT for sentiment classification**, achieving 92% accuracy and 0.91 macro-F1 on a held-out evaluation dataset.
- Deployed a **containerized ML application using Streamlit, Docker and AWS S3**, achieving 300 ms p95 inference latency while supporting 130 concurrent users.

WORK EXPERIENCE

Siemens | Software Engineer Intern | Remote Aug 2024 - Jul 2025

- Engineered 2 **AI-powered applications using Node.js, PostgreSQL and integrated AI services**, supporting 2,500+ users with 99.5% application availability.
- Improved chatbot retrieval relevance by 40% through a **Pinecone-backed RAG pipeline**, using semantic search to support 1,200+ conversations.
- Optimized **AI service integration and API performance**, achieving 850 ms p95 response latency across 10,000 benchmark queries.
- **Led 3 engineering interns** through AI feature development and API integration, resolving 9 production issues through debugging, log analysis and testing.

Unisys | Software Engineering Intern | Remote Apr 2024 - Jun 2024

- Instrumented **checkout funnels and user-event tracking** for a marketplace serving 8,000+ users and processing 1,600+ orders, generating $20K in gross merchandise value.
- Increased checkout conversion by **22%**, from 41% to 50%, through funnel analysis and one-click checkout improvements, contributing $14K in revenue.
- Engineered **WebSocket-based event delivery** for buyer-seller messaging, supporting 350 concurrent connections with sub-150 ms latency and 99.8% message delivery.
- Improved mobile conversion by **8%** through Lighthouse-guided performance analysis, reducing page load time to under 1.2 seconds.

SKILLS

Languages: Python, SQL, Java, C++, Go, TypeScript, JavaScript
ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, DeepSpeed, Transformers, NLP, Feature Engineering, Model Evaluation
Data Engineering:: Pandas, NumPy, PostgreSQL, Kafka, Redis, MongoDB, Data Pipelines, Stream Processing, SQL Optimization
ML Systems & MLOps: Distributed Training, Model Inference, Docker, Kubernetes, CI/CD, GitHub Actions, Experimentation
Cloud & Tools: AWS (EC2, S3, Lambda), Azure, Streamlit, Prometheus, Grafana, Git

CERTIFICATIONS & ACHIEVEMENTS

- Open Source - Merged PR #472 into mindsdb/anton, fixing a bug that deleted saved user rules and adding 3 regression tests. Sep 2026
- Publication - "Detection of DDoS Attacks in Fog Computing using Interpretable Quantum-based 2Q Learning Approach" - IEEE Jan 2025